

9708/31/0/n/18

1-A:

A – This is correct because “external cost” is the difference between social cost and private cost.

B – Private cost is the difference between social cost and external cost.

C and D: This is incorrect because social cost is the sum of private cost and external cost.

2-D:

A – A “new road” built by for example: a firm, when results in an increase in sales of other shops in the town center, represents “positive externality from production”

B – A successful promotion of healthy eating, when reduces days lost from work, results in an increase in output at the workplace, which in turn represents “positive externality from consumption”

C – Discharge of waste from a factory that damages fish in a local river, represents “negative externality from production”

D – Increased popularity of smoking that inflates the national health bill, or in other words, increases the demand for healthcare, as not only the smoker, but also, the passive smoker, will be facing health issues too, represents a negative externality from consumption. Therefore, this becomes correct.

3-A:

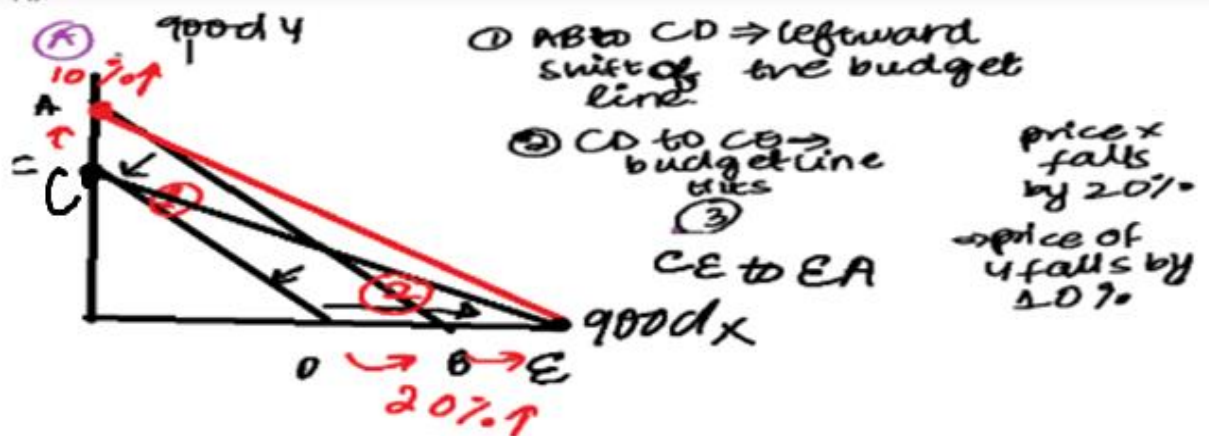
A – This is correct because allocative efficiency occurs when marginal cost pricing is done or in other words, marginal cost is equal to price.

B – Dynamic efficiency occurs when Average cost at the same level of output decreases over time due to re-investment of supernormal profits in research and development.

C – Productive efficiency occurs in the long run when the Average total cost is minimized or in other words, $MC=ATC$.

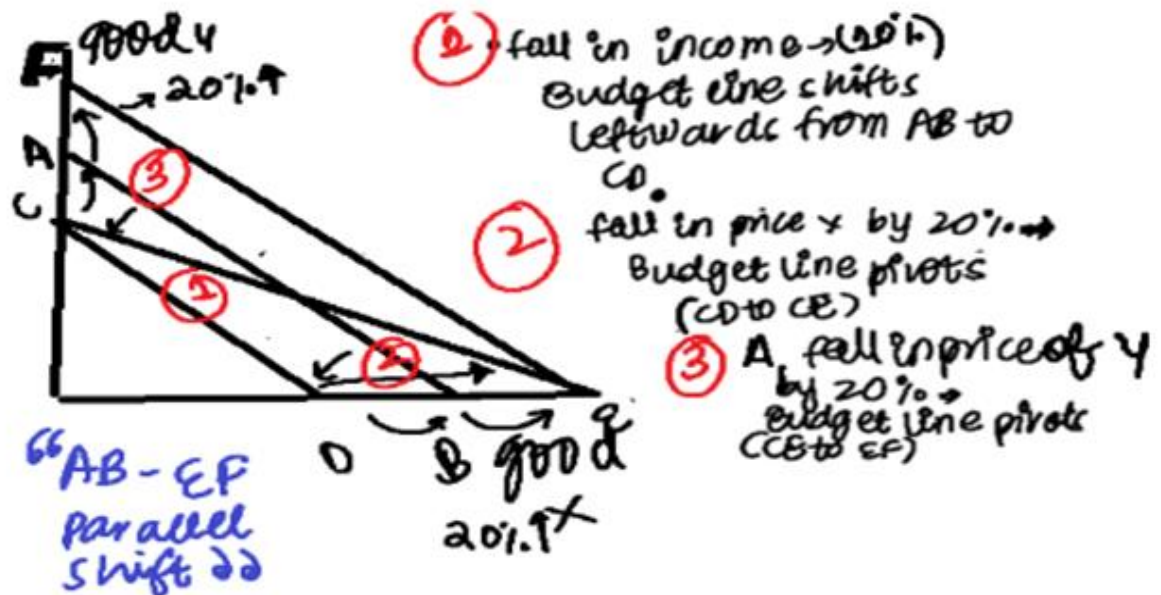
D: Technical efficiency refers to the ability of a firm or organization to produce a given output with the least amount of inputs (such as labor, capital, materials, etc.) necessary, given the current state of technology and resources.

4-B:

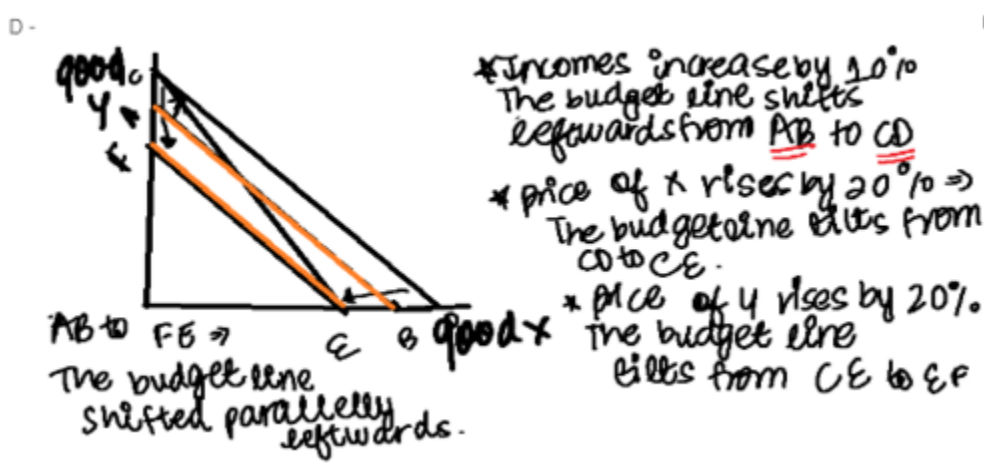
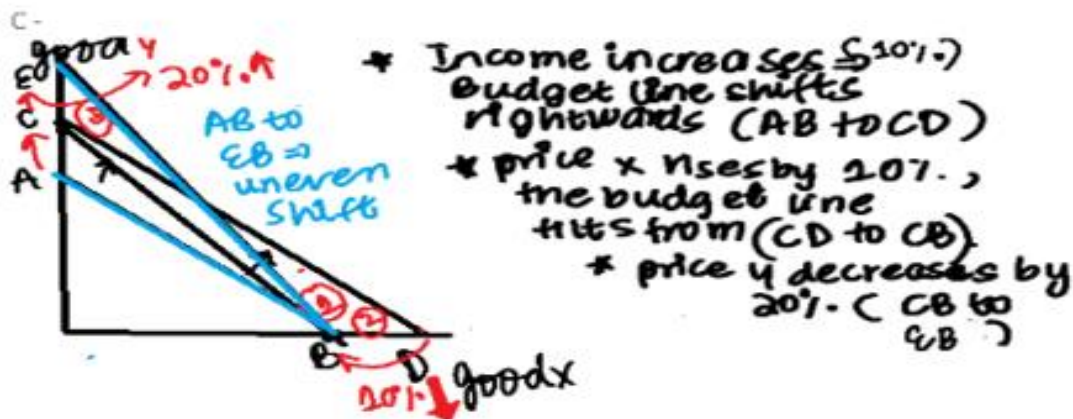


The shift is not parallel.

B -



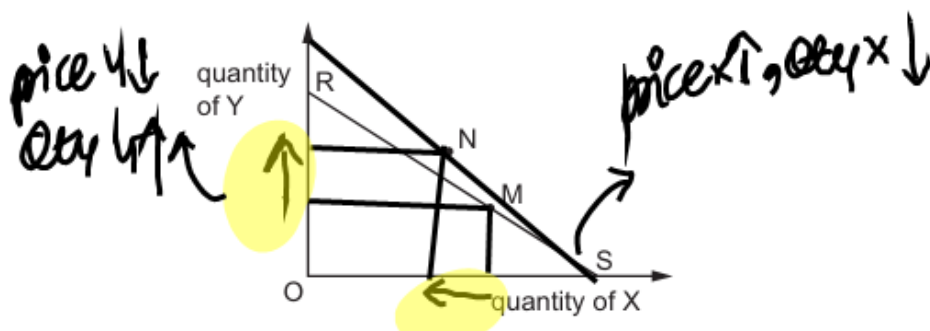
The budget line has shifted parallelly rightwards from "AB to EF", which is why this is



5-B:

3

The line RS in the diagram shows the different combinations of goods X and Y that a consumer can afford with her present income.



A – A change in tastes or preferences will result in a movement along the indifference curve.

B – This is correct because a fall in the price of good Y resulted in an extension in qty consumed of good Y and also, a rise in the price of good X resulted in a contraction in qty consumed of X. This explains why key B is correct.

C – If the prices of both the goods increased, then the quantity of both goods may have reduced.

D – Equal percentage increases in income and in both prices may have left the budget line unchanged.

6-C:

The profit maximizing level of output is where $MC=MR$. This can be seen in the range of 5-6 tonnes.

6 The table shows the total revenue and marginal cost of a firm at different levels of production.

	production (tonnes)	total revenue (\$)	marginal cost (\$)
	2	70	
	3	120	15
	4	160	20
	5	190	25
	6	220	30
	7	240	40

Within which output range will the firm's profits be maximised? $\rightarrow MC = MR$

- A 2–3 tonnes
- B 3–4 tonnes
- C 5–6 tonnes**
- D 6–7 tonnes

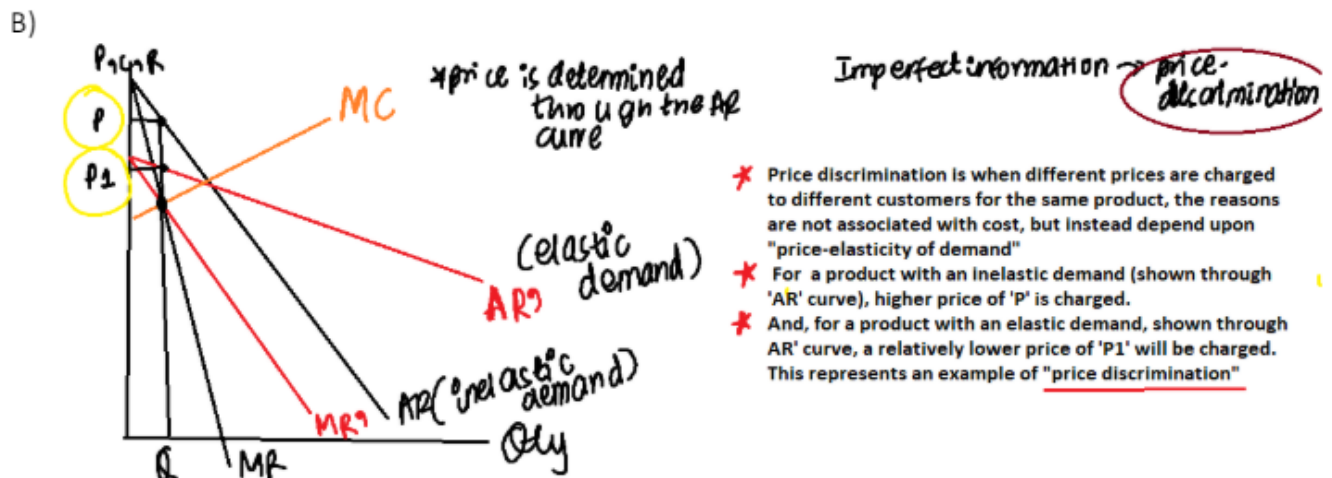
7-B:

Initially, when the firm operated with an objective of “profit maximization”, it produced a profit maximizing level of output, which is obtained from the equilibrium where the Marginal cost is equal to the Marginal Revenue($MC=MR$), which is “OW”

Now, when it charges a price, also known as Average Revenue(AR), equal to the marginal cost, this explains a situation whereby “allocative efficiency is achieved”

Here, the output of "OY" will be produced. This means that the increase in output will be by "WY" This is why, key B, becomes correct.

8-B:



B – Price elasticities of demand **"must"** be different for "price discrimination to be profitable", based on the definition of "price-discrimination", as can be seen above.

A, C, D : Levels of income(option A), the ages of the consumers(option C), the times of consumption(option D), do help in forming different market segments, based on which, a firm may charge different prices(as with **"third-degree price discrimination"**). But it is not necessary or **"must"** for any of the feature to be present to be able to do "price-discrimination"

9-B:

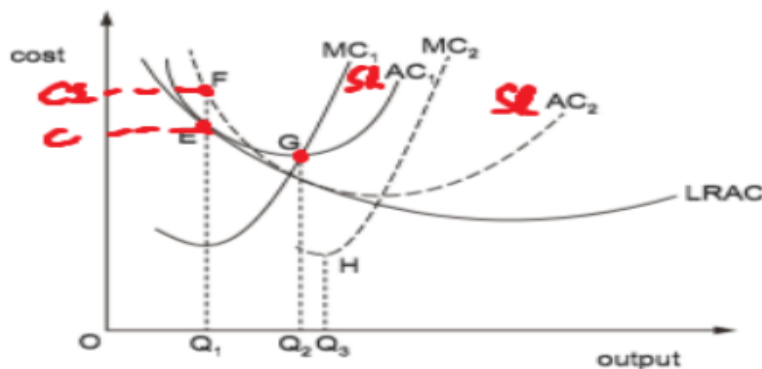
A) Merging with another computer manufacturer would be an example of horizontal integration, not forward vertical integration. This is because both companies are at the same stage of the supply chain, producing similar products. Horizontal integration aims to increase market share, reduce competition, and achieve economies of scale.

B) Merging with a computer retailer is an example of forward vertical integration. The computer manufacturer is acquiring a business that is closer to the end customer, gaining control over the distribution and sales of their product.

C) Merging with a silicon chip manufacturer would be an example of backward vertical integration, not forward vertical integration.

D) Merging with a software developer is an example of related diversification or conglomeration, not forward vertical integration. This is because software development is a different business from computer manufacturing, although the two products are related.

10-A:



A - At output Q_1 , the firm has the option to either operate on the plant curve or the short-run average cost curve "AC1" or "AC2"

It will choose to operate on point E because E generates a lower average cost than F. And, at this level of output, costs cannot be further minimized. This is why, point E represents a "productively-efficient position"

B- At output Q_1 , point F is not preferred to point E, this is because F represents a higher average cost than E.

C - In order to be productively efficient in the long-run, the short-run average cost and the long-run average cost must be minimized. Here, the short-run average cost is being minimized since Marginal cost curve is intersecting AC_1 . But, the long-run average cost curve is not being minimized since Long-Run marginal cost (LRMC) is not tangential to Long-run Average Cost (LRAC)

D - We only have information about costs, and based on this, we cannot comment on profits.

11-B:

A cartel is a group of firms that collude to **restrict output and raise prices**.

B) When new firms enter the market, it becomes harder for the cartel to maintain its control over output and prices. These new firms may not be part of the cartel agreement and this may cause an increase in overall market output and a decrease in prices. This can cause tension among cartel members and lead to the cartel's breakdown.

A) An agreement that the cartel produces a homogenous product actually helps the cartel maintain its control, as it reduces differentiation and makes it easier to coordinate output.

C) Stable patterns of demand for the good can actually help the cartel maintain its control, as it makes it easier to predict and coordinate output.

D) Trademarks, acting as a barrier to entry to the market, would actually help the cartel maintain its control by reducing the likelihood of new firms entering the market.

12-B:

A: An increase in costs of borrowing from commercial banks will increase the costs of small firms, making their survival more difficult.

B : An increase in personal services as a proportion of GDP, represents a rise in consumption or spending on personal services, and a rising demand for them will make it easier for small firms to survive as small firms mainly rely on personal services.

C: An increase in the rate of change of information technology will increase the need for higher investment, making it difficult for small firms to survive.

D: An increase in the ratio of capital to labor employed will increase the need for higher investment in capital equipment, as a result of which, small firms may find it difficult to survive

13-D:

→ In the case of negative externalities (external costs), internalizing the externality involves making the party responsible for the externality bear the cost of the harm they are causing. This can be done through taxes or regulations that increase the private cost of the action to reflect its social cost. This, in turn, encourages the reduction of activities that generate negative externalities.

A - Closing the part of the mine and importing coal does not directly internalize the external cost associated with the pollution created by the mining operations.

B - This option targets the miners' wages, putting them to a disadvantage, rather than the company's operations or output. It also doesn't directly address the environmental externality or the company's role in generating it.

Internalizing the externality involves making the party responsible for the externality bear the cost of the harm they are causing, which in this case, is “the private company”

C – If the government pays for the external costs of restoring the environment, this means that, external costs will keep on increasing. Here also, externalities are not internalized. Internalizing the externality involves making the party responsible for the externality bear the cost of the harm they are causing, which in this case, is “the private company”

D) Placing a tax equivalent to the external cost on the output the company produces is the correct option for internalizing the externality created by the coal mining company's operations.

-> **Tax on Output: Placing a tax on the output the company produces is a direct way to internalize the external costs associated with the pollution caused by the coal mine. This tax is known as a Pigouvian tax, which aims to align private costs with social costs.**

-> **Internalizing Externality: By imposing a tax on the output, the government makes the company bear the cost of the negative externalities it is creating. This incentivizes the company to account for the environmental damage and incorporate it into its decision-making process. As a result, the company is likely to reduce its output, as the additional cost from the tax makes some mining activities less profitable**

14-C:

→ Using pollution permits incorporates the cost of the pollution into the production costs and helps overcome market failure.

A - This statement is not accurate. Tradable permits can be **bought and sold among firms**, as less-polluting firms can sell some of their permits to more polluting firms

B - There is still a need for a system to verify that companies are correctly reporting their emissions and trading activities to ensure the integrity of the system, which means that **enforcement costs are still being incurred:**

→ If a company holds a permit allowing it to emit 100 kg of carbon and it emits 150 kg without buying additional permits, it would likely be in **violation of the permit system's regulations. These penalties could include fines, sanctions, or restrictions on the company's future emissions.**

→ **Regulatory authorities responsible for overseeing the permit system would likely take enforcement actions against the company for exceeding its emissions limit. This could include investigating the violation, imposing fines, and ensuring that the company takes corrective actions to comply with the regulations.**

C - Tradable pollution permits create a **market for pollution rights, allowing firms to buy and sell permits based on their pollution needs.** This mechanism introduces market dynamics into the process of pollution control. Companies with lower pollution abatement costs will have an incentive to reduce emissions and then sell their unused permits to companies with higher abatement costs. This is why, key C, becomes correct.

The existence of “pollution rights market” is shown below:



D - Tradable permits do not guarantee a zero level of pollution. They aim to **reduce pollution** by setting limits on emissions and allowing firms to trade permits, but **the total level of pollution will depend on the number of permits issued**.

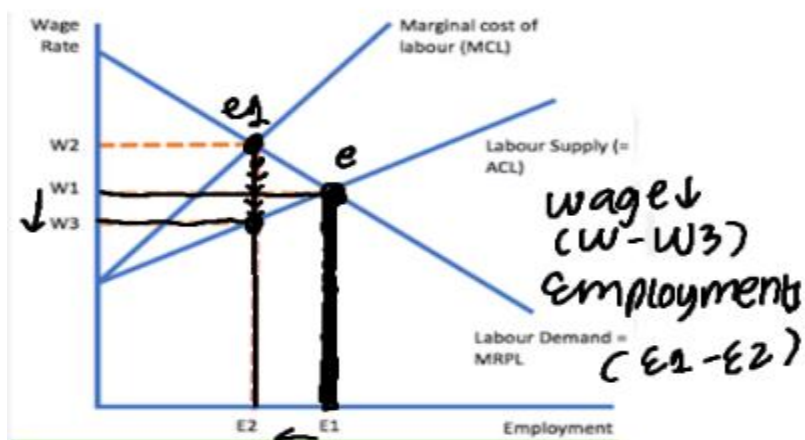
15-D:

A – An industry protected by tariffs on imports represents “protectionism”, which will not justify government intervention to raise the wages of a group of workers.

B – In an industry where a trade union can negotiate wages itself because of its collective bargaining power, government intervention to raise the wages of a group of workers will not be required.

C – In an industry where the output is produced by a single firm monopolist, this represents the existence of “monopoly power” in a product market.

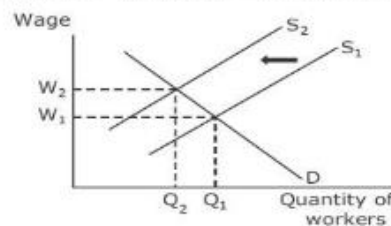
D - A monopsony is a single buyer of labor, which will exert its exploitative behavior and reduce both, the wage level and employment, as can be seen in the diagram below. Therefore, in such a case, government intervention to raise the wages of a group of workers will be justified.



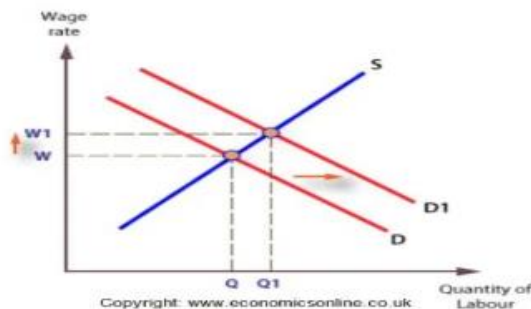
16-D:

- A decrease in the supply of female workers will result in a **shortage** of female workers, which will exert an **upward pressure on the “wage rate”**, as can be seen in the diagram below.

A Decrease in the Supply of Workers in the Labor Market



- This will lead to **employers increasing the demand for male workers**, and since this **rise in demand for labor results in a shortage of male workers**, it will exert an **upward pressure on the “wage rate”**, as can be seen in the diagram below.



This is why, key D, becomes correct.

17-B:

Claimant count considered those people as unemployed who were officially registered as unemployed and were claiming unemployment benefits.

International labor force survey overcame this problem and considered those people as unemployed who **have been actively seeking employment at the current wage rate and are available to start work within the next two weeks**. This is why, only key B is correct.

18-C:

$$TC_{11} - TC_{10} = MC_{11}$$
$$[3.25 \times 11] - [3 \times 10] \Rightarrow 5.75$$

19-C:

C - When output increases at a faster rate than labour earnings, labour costs per unit of output decrease. This is because the fixed amount of labour earnings is now spread over a larger quantity of output, reducing the labour cost per unit.

A) An increase in average earnings that is less than the increase in employment would actually lead to an increase in labour costs per unit of output, as the total labour earnings increase at a faster rate than output due to a higher increase in employment.

B) An increase in labour productivity that is less than the increase in average earnings would also lead to an increase in labour costs per unit of output, as the productivity gain is offset by the larger increase in earnings.

D) An increase in wage rates that is greater than the increase in output per worker would lead to a significant increase in labour costs per unit of output, as the higher wage rates outpace the productivity gain.

20-A:

Formula of multiplier $\Rightarrow \frac{1}{\frac{mps + mpt}{mpm} + 1}$ or $\frac{1}{1 - mpc}$

$$\frac{1}{0.10 + 0.25 + 1} \rightarrow \frac{1}{0.5} \rightarrow 2.00$$

21-B:

The Human Development Index (HDI) is a composite statistic that combines three dimensions:

1. Life expectancy at birth (health)
2. Expected years of schooling (education)
3. Gross National Income (GNI) per capita (standard of living)

The HDI values range from 0 (low) to 1 (high), with higher values indicating better human development outcomes. The HDI is used to rank countries and assess their progress in achieving human development goals.

This explains why key B is correct.

22-D:

If multinational companies train the workforce (B), bring improved technology (A), this will be most likely to increase the productive potential of the economy. Similarly, they also bring benefits in form of increased tax revenues for the government (C).

However, when they provide tough competition, they may outcompete small scale domestic producers, which may harm domestic industry (D). As a result of which, this becomes correct.

23-C:

A – If the government raises corporation taxes as part of contractionary-fiscal policy, it will have an effect of further dampening Aggregate demand, thereby increasing cyclical unemployment.

B – Increasing the exchange rate will increase the prices of exported goods and will decrease the prices of imported goods, which will lead to a Balance of trade deficit in the long run. As a part of contractionary monetary policy, it will have an effect of further dampening Aggregate demand, thereby increasing cyclical unemployment.

C – A budget surplus is a situation in which tax revenues exceed government expenditures. However, when a budget surplus is reduced, this represents that tax revenues are falling and government expenditures are rising as part of expansionary fiscal policy. This will have an effect of boosting Aggregate demand that will reduce demand deficient unemployment.

D – If money supply is reduced, this will bring about an increase in interest rates as part of a contractionary monetary policy. After having raised costs of borrowing and dampening investment expenditure, this will bring about a reduction in Aggregate demand, thus further exacerbating the levels of cyclical unemployment.

24-C:

A – A decrease in imports represents a fall in marginal propensity to import, which since will increase the marginal propensity to consume, will increase the size of the multiplier.

B – A decrease in the rate of interest after having brought about a reduction in the marginal propensity to save, will result in an increase in the marginal propensity to consume and because of this, the value of the multiplier will rise.

C – An increase in income taxes represent a rise in marginal propensity to tax. This will bring about a fall in marginal propensity to consume. As a result of which, the value of the multiplier will fall. This makes option C correct.

D – Again as mentioned above, an increase in the marginal propensity to consume will boost the value of the multiplier.

25-B

A reduction in interest rates after having reduced the incentives to save, will stimulate an increase in spending, as a result of which, Aggregate demand will increase. This represents actual economic growth or an increase in economic growth.

As a result of this increase in Aggregate demand, there will be a rise in production and demand for labor which will subsequently decrease unemployment.

A - This can be seen in the figure above that an inflationary gap exists when Aggregate demand is greater than full employment level of income.

B – When leakages exceed injections at full employment level of income, there exists a deflationary gap in the economy.

C – When money wages rise faster than labor productivity, this can raise the costs of production of firms, thereby reducing AS, in turn leading to cost push inflationary pressures,

D – A faster increase in money supply can again lead to demand pull inflation, but it may not necessarily translate in to an inflationary gap.

28-C:

The Accelerator Theory posits that investment (I) is a function of the rate of change in consumer demand (ΔC). **The basic assumption is that businesses invest more when consumer demand is increasing rapidly, as they need to expand capacity to meet growing demand.**

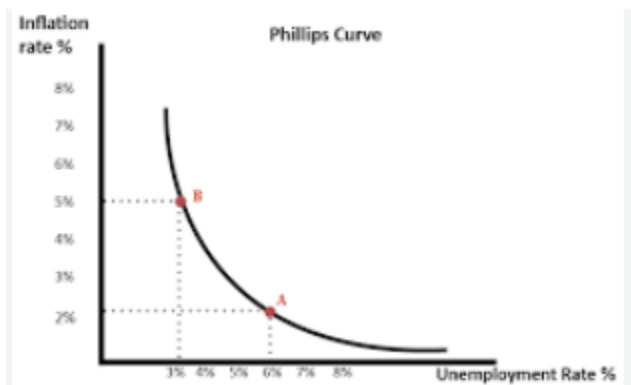
Here's why the other options are incorrect:

A) Business confidence is not the primary driver of investment in the Accelerator Theory.

B) Savings are not directly related to investment decisions in this theory.

D) Interest rates are not a key factor in the Accelerator Theory.

29-A ; A typical Philips curve shows a negative relationship between inflation and unemployment.



The Phillips Curve represents this trade-off between inflation and unemployment, showing that:

- As unemployment falls (improving labor market), inflation tends to rise (as wages and production costs increase).
- As unemployment rises (worsening labor market), inflation tends to fall (as wages and production costs decrease).

The table shows data on the unemployment and inflation rates for four countries in 2010 and 2014.

country	unemployment rate (% of workforce)		inflation rate (% per annum)	
	2010	2014	2010	2014
Australia	5.2	6.1 ↑	2.7	1.6 ↓
Canada	8.0	6.9 ↓	2.2	1.9 ↓
Japan	5.1	3.6 ↓	-0.3	2.5 ↑
United States	9.6	6.2 ↓	1.7	0.6 ↓

Which countries had a typical Phillips curve?

- ✓ A Australia and Japan
 B Australia only
 C Canada and United States
 D Japan only

↳ -ve relationship

30-D:

A – A decrease in the rate of interest after having reduced the incentives to save, and subsequently savings, since will also bring about an increase in consumption of imported goods, means that the import expenditure will increase, which can in-turn help to reduce a bot surplus. But since, domestic consumption may also rise because of this, there will be an increase in Aggregate demand, and when Aggregate demand outpaces Aggregate supply, there will be shortages that will translate into demand pull inflation.

B – Devaluation of the currency will further lead to an increase in the bot surplus in the long run when marshall Lerner condition is held, that is when the combined sum of elasticities of demand of imports and exports is greater than 1. Additonally, when because of these net injections in to the circular flow, there will be an increase in Aggregate demand, there will come a point when Aggregate demand outpaces Aggregate supply, and so, there will be shortages that will translate into demand pull inflation.

C – An increase in direct taxation results in a decrease in disposable incomes, as a result of which, consumption expenditure will decrease. This will bring about a fall in Aggregate demand that will reduce demand pull inflation. But since, consumption of imported goods will fall too, this will translate into an increase in BOT surplus.

D – If I remove tariffs on imports, this will make imported goods cheaper, and when because of this, import expenditures rise, the BOT surplus will reduce. Apart from this, also because of consumers switching to imported goods, when the domestic consumption falls, there is a decrease in Aggregate demand that reduces demand pull inflation in the economy. This is why, key D, becomes correct.